

YUBILEYNAYA, Russian Federation

WMO: 219310

Lat: 70.767N

Lon: 136.217E

Elev: 80

StdP: 14.65

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-50.7	-47.5	-53.7	0.2	-50.1	-50.9	0.3	-47.3	24.7	-19.6	20.4	-6.8	4.2	210	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.5	76.4	60.8	71.6	58.6	67.0	56.2	63.1	73.2	60.2	69.2	57.3	65.7	6.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.8	74.2	67.1	55.7	66.3	64.1	52.9	59.6	61.6	28.7	73.0	26.6	69.4	24.7	65.8	72.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.3	19.4	16.4	DB	-52.8	83.6	2.8	4.3	-54.8	86.7	-56.5	89.3	-58.1	91.7	-60.1	94.8	(4)
(5)			WB	-52.8	66.2	2.8	2.5	-54.8	68.0	-56.5	69.4	-58.1	70.8	-60.1	72.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	-29.0	-27.7	-12.5	6.0	25.5	47.2	52.5	50.9	37.2	12.8	-10.3	-26.0	(6)
(7)		DBStd	32.25	11.70	12.36	14.59	12.36	11.30	9.99	9.05	6.76	6.62	12.57	11.90	11.41	(7)
(8)		HDD50	14668	2450	2176	1938	1319	761	171	78	71	388	1153	1808	2355	(8)
(9)		HDD65	19820	2915	2596	2403	1769	1224	542	402	439	834	1618	2258	2820	(9)
(10)		CDD50	345	0	0	0	0	2	86	155	98	4	0	0	0	(10)
(11)		CDD65	23	0	0	0	0	0	7	15	1	0	0	0	0	(11)
(12)		CDH74	244	0	0	0	0	0	77	148	19	0	0	0	0	(12)
(13)		CDH80	50	0	0	0	0	0	11	39	0	0	0	0	0	(13)
(14)	Wind	WSAvg	6.9	5.5	5.2	5.8	6.3	7.9	9.1	8.5	7.6	6.7	6.7	7.4	5.8	(14)
(15)	Precipitation	PrecAvg	9.6	0.3	0.4	0.3	0.3	0.7	1.1	2.1	1.6	1.1	0.9	0.6	0.4	(15)
(16)		PrecMax	17.9	0.6	1.2	0.6	0.6	1.4	2.5	4.9	4.6	2.1	4.5	1.1	1.1	(16)
(17)		PrecMin	7.2	0.1	0.0	0.1	0.2	0.1	0.0	0.6	0.6	0.1	0.1	0.1	0.1	(17)
(18)		PrecStd	2.5	0.2	0.3	0.1	0.1	0.4	0.6	1.3	0.9	0.5	0.9	0.3	0.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.4	10.2	27.3	39.8	62.7	81.6	84.6	76.8	63.1	33.8	21.6	6.3	(19)
(20)			MCWB	2.3	8.7	24.9	32.8	49.9	61.8	65.1	60.3	50.4	30.5	19.7	5.4	(20)
(21)		2%	DB	-0.8	3.5	22.4	34.8	53.2	75.5	77.8	72.0	56.5	32.1	16.7	0.7	(21)
(22)			MCWB	-1.4	2.7	20.0	29.4	43.8	58.7	62.5	58.6	47.0	30.0	16.0	0.0	(22)
(23)		5%	DB	-5.5	-4.5	16.9	30.2	46.0	70.0	72.2	67.5	51.1	30.5	12.0	-4.1	(23)
(24)			MCWB	-5.9	-5.2	15.1	25.7	39.0	56.2	60.1	57.1	45.1	28.6	11.0	-4.8	(24)
(25)		10%	DB	-11.6	-11.0	10.2	25.1	41.2	64.3	67.6	63.1	46.8	28.6	7.7	-8.5	(25)
(26)			MCWB	-11.9	-11.5	8.4	21.8	36.4	53.7	57.3	55.4	42.4	26.7	6.9	-8.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.7	8.5	25.1	33.4	50.4	63.4	67.2	64.4	51.3	32.1	20.1	5.6	(27)
(28)			MCD B	3.3	10.1	26.9	38.6	61.9	77.1	79.3	71.6	60.0	33.0	21.5	6.3	(28)
(29)		2%	WB	-1.3	2.7	20.0	29.9	44.1	60.6	64.5	61.3	48.6	30.3	15.8	0.1	(29)
(30)			MCD B	-0.9	3.6	21.9	34.9	52.6	73.0	74.7	68.7	55.0	31.5	16.7	0.6	(30)
(31)		5%	WB	-6.0	-4.9	15.0	25.8	39.5	57.2	61.3	58.7	45.5	28.8	11.0	-4.6	(31)
(32)			MCD B	-5.7	-4.5	17.0	30.1	45.0	69.1	70.8	64.9	49.8	30.4	11.9	-4.0	(32)
(33)		10%	WB	-12.1	-11.4	8.4	21.7	36.4	53.9	58.0	55.9	42.8	26.7	7.0	-8.9	(33)
(34)			MCD B	-11.8	-11.1	10.1	25.0	40.8	63.4	66.2	61.8	46.9	28.6	7.8	-8.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.3	11.1	15.1	17.3	12.6	15.2	14.5	13.8	9.8	9.4	10.9	10.2	(35)
(36)			MCDBR	12.8	17.2	16.8	20.7	19.3	24.2	22.3	22.3	17.5	7.4	13.8	13.2	(36)
(37)			MCWBR	12.4	16.7	15.5	16.5	10.8	11.8	10.5	11.6	9.9	6.7	13.5	13.0	(37)
(38)		5% WB	MCDBR	12.7	17.2	16.8	20.5	17.8	22.3	21.1	19.5	16.0	6.6	13.6	13.1	(38)
(39)			MCWBR	12.3	16.7	15.4	16.5	10.1	11.6	11.4	11.8	9.9	6.3	13.1	12.9	(39)
(40)	Clear-Sky Solar Irradiance	taub		N/A	0.203	0.279	0.317	0.354	0.352	0.408	0.375	0.297	0.217	N/A	N/A	(40)
(41)		taud		N/A	2.004	2.062	2.062	2.167	2.420	2.263	2.339	2.466	2.082	N/A	N/A	(41)
(42)		Ebn at Noon		0	159	220	252	258	265	240	234	227	146	0	0	(42)
(43)		Edh at Noon		0	11	25	36	37	30	34	28	18	10	0	0	(43)
(44)	All-Sky Solar Radiation	RadAvg		0	102	600	1386	1890	1761	1500	1019	508	171	0	0	(44)
(45)		RadStd		0	9	24	46	102	148	98	115	40	22	0	0	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page